

Aayush Raj Pandey

Boise, Idaho | pandeyaayushraj@gmail.com | GitHub & Portfolio Available Upon Request

Education

The College of Idaho

Bachelor's in Physics and Mathematics | Expected Graduation: 2027

Relevant Coursework: Statistical Machine Learning, Data Analysis, Physics Modeling, Linear Algebra, Differential Equations

Technical Skills

Programming: Python, pandas, NumPy, scikit-learn, matplotlib, Jupyter Notebook

Machine Learning & Analytics: Logistic Regression, Random Forest, XGBoost, Feature Engineering, Cross-Validation, ROC-AUC Evaluation

Tools & Technologies: GitHub, VSCode, Quarto, Excel, LaTeX

Relevant Experience

Founder & Partner | Evolption — Remote/New York | 2025 – Present

- Co-founded a cosmetics business collaborating with more than 100 medical spas across New York.
- Led partnership negotiations, operations, client acquisition, and strategic expansion planning.
- Generated approximately \$10,000 profit within the first few months of operation.

HR Assistant — The College of Idaho | 2024 – Present

- Maintain employee records and support HR documentation and administrative workflows.
- Collaborate with staff to improve organization and operational efficiency.

Events Associate — The College of Idaho | 2024 – Present

- Coordinate logistics, scheduling, lighting, and sound operations for campus-wide events.

Projects

NBA Win Prediction Modeling | Python, scikit-learn, pandas

- Built predictive ML pipelines using modern NBA datasets from 2021–2025.
- Engineered rolling matchup differential features including offensive, defensive, and win percentage trends.
- Compared Logistic Regression, Ridge, Lasso, Random Forest, and XGBoost using temporal validation.
- Improved predictive performance primarily through feature engineering and ROC-AUC optimization.

MH370 Search Physics Modeling Project | Python, Bayesian Modeling

- Developed a physics-based debris search model using Bayesian probability and ocean drift simulations.
- Applied mathematical modeling and ML concepts to aviation search optimization.

Leadership & Activities

Student Senator — The College of Idaho Student Senate | **President** — Soccer Club | **Founder** — Discite Educational Initiative | **ISO Executive Council** | **Annotator** — Sigma Chi

Languages

English (Fluent) | Nepali (Fluent) | Spanish (Intermediate)